CV

Gowrisha C. Vishwa Kumar



Senior Cloud & AI Platform Architect | LLM & RAG | Kubernetes | Terraform | Azure

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Professional Summary

Cloud and Platform Architect with 14+ years of IT experience and 7+ years in cloud, combining enterprise Azure architecture, AKS/Kubernetes, Terraform and CI/CD with 3+ years inside Santander's regulated banking environment. Experienced in designing secure, automated cloud foundations under DORA, BaFin and GDPR constraints, and now applying that foundation hands-on to LLM, RAG and AI platform architecture: model APIs, embeddings, retrieval, vector search, identity, private networking, observability and cost governance. Brings regulated-cloud experience to AI architecture decisions involving data access, identity, model governance, auditability and secure deployment, and translates architecture into working technical solutions.

Professional Experience

Technological Architect II, Infrastructure & Integration

Sep 2023 - Present

Santander Consumer Technology Services GmbH | Mönchengladbach, Germany

- Design secure Azure cloud foundations for regulated workloads: CAF-based Landing Zones, management groups, subscriptions and centralised platform services
- Convert architecture standards into reusable Terraform modules and CI/CD controls in GitHub Actions and Azure DevOps, with Azure Policy and RBAC guardrails, reducing manual provisioning effort by approximately 40%
- Design and engineer AKS platform architecture: cluster networking, workload identity, security controls and Azure service integration for application teams
- Design identity and secret patterns with Entra ID, Managed Identity, workload identity (OIDC) and Key Vault, reducing reliance on stored credentials in pipelines and services
- Design private networking for cloud-native platforms: hub-and-spoke, Azure Firewall, Private Endpoints, Private DNS, NSGs, VPN and ExpressRoute
- Apply governance and auditability controls in a DORA, BaFin and GDPR context through Azure Policy, PIM and Defender for Cloud
- Implement observability with Azure Monitor, Log Analytics, Application Insights and Dynatrace, and support platform and application teams in troubleshooting
- Work across security, networking, platform and application teams on architecture reviews and technical decisions
- Contributed to approximately 25% cloud cost reduction through governance, resource optimisation and FinOps practices

Technical Lead

Aug 2022 - May 2023

Sopra Steria | India

- Worked with AKS workloads, deployment and networking; designed Entra ID, Managed Identity and OIDC workload identity patterns for keyless authentication
- Automated Terraform provisioning and application delivery through CI/CD pipelines with security and governance controls

Associate Technical Lead

Oct 2021 - Jul 2022

Tech Mahindra (via Brainscale acquisition) | India

- Implemented Kubernetes workloads on AKS and GKE; built Terraform and CI/CD automation for repeatable provisioning across Azure and GCP
- Designed security and disaster recovery solutions across Azure and GCP

Cloud Engineer

Oct 2017 - Oct 2021

Brainscale Inc. (acquired by Tech Mahindra) | India

- Engineered a 190 TB AWS S3 to Azure Blob Storage migration with zero data loss, including AWS IAM policy mapping and staged validation
- Delivered migration of 50+ enterprise workloads to Azure, reducing infrastructure costs by 30%
- Designed disaster recovery and backup strategies that reduced recovery time objective (RTO) from 12 hours to 4 hours

Systems Engineer

Jun 2015 - Oct 2017

NetApp Inc. | India

- Administered VMware vSphere, ESXi and vCenter for enterprise Windows workloads, including VM lifecycle, patching, monitoring and incident resolution

AI / LLM Engineering Projects

Self-directed engineering work outside Santander employment. Not production systems.

GreenOps AI Platform - LLM, RAG and Secure AI Architecture

Portfolio / Engineering Lab

Azure OpenAI | Azure AI Foundry | Azure AI Search | AKS | Terraform | GitHub Actions

- Designed an AKS-based generative AI platform architecture integrating Azure OpenAI and Azure AI Foundry model endpoints with enterprise identity, private networking and observability controls, including model routing across managed and self-hosted options
- Designed a RAG flow covering document ingestion, chunking, embeddings, vector indexing in Azure AI Search, retrieval, context augmentation and grounded response generation to reduce hallucination risk
- Applied Kubernetes workload identity (OIDC), Key Vault and private connectivity to remove stored credentials and keep model and data traffic inside controlled trust boundaries
- Built the delivery model with Terraform and GitHub Actions, including Checkov infrastructure validation, Trivy container scanning, Syft SBOM generation and Cosign / Notation image-signing concepts
- Evaluated managed versus self-hosted LLM deployment options across security, regulatory fit, operational complexity, latency and cost to support enterprise architecture decisions.
- Included Hugging Face self-hosted model options in the design, and defined an observability approach using OpenTelemetry, Prometheus and Grafana, with FinOps cost considerations

Local LLM Engineering and API Integration

Hands-on Lab

LM Studio | Ollama | Docker | REST APIs

- Ran and tested local LLMs with LM Studio and Ollama, exposing model endpoints through local APIs for application integration experiments
- Validated LLM REST API calls from host and containerised (Docker) environments, working through container-to-host connectivity for self-hosted inference
- Compared local model execution with managed AI platform patterns on security, latency, infrastructure and cost

> **AI Architecture Approach**
> User / Application -> API / AI Orchestrator -> Identity and Authorization -> Retrieval -> Embeddings -> Vector Index -> Context Augmentation -> LLM -> Guardrails -> Logging, Evaluation and Monitoring. Each layer is designed against identity, networking, data access and privacy, model selection, retrieval quality, hallucination risk, security, compliance, latency, scalability and cost.

Technical Skills

AI & Generative AI	LLMs, Generative AI, RAG, Embeddings, Vector Search, Prompt Engineering, Grounding, Model Selection, Agentic AI Concepts, AI Platform Architecture, Azure OpenAI, Azure AI Foundry, Azure AI Search, Local LLMs
AI Operations (Concepts)	LLMOps, AI Evaluation, Model Versioning, AI Observability, AI Security, Responsible AI
Cloud & Platform	Microsoft Azure, Azure Landing Zones, AKS, Kubernetes, Containers, Docker, Model Endpoint Architecture, GCP, GKE
Infrastructure as Code	Terraform, Terraform Modules, Bicep, Infrastructure Automation
DevOps & Engineering	GitHub Actions, Azure DevOps, CI/CD, Git, GitHub, REST APIs, Python (working knowledge - automation and AI experimentation)
Security & Governance	Entra ID, RBAC, PIM, Managed Identity, Workload Identity (OIDC), Key Vault, Azure Policy, Defender for Cloud, Private Endpoints, Zero Trust, DORA, BaFin, GDPR
Networking	Hub-Spoke, Private Endpoints, Private DNS, Azure Firewall, NSGs, VPN, ExpressRoute
Observability	Azure Monitor, Log Analytics, Application Insights, Dynatrace, OpenTelemetry (concepts)
FinOps	Cost Optimization, Resource Governance, Cloud Cost Management
Additional Cloud Exposure	AWS S3 and AWS IAM (cross-cloud migration to Azure)

Certifications

Microsoft Certified: Azure Solutions Architect Expert, Microsoft Certified: DevOps Engineer Expert, HashiCorp Certified: Terraform Associate, Microsoft Certified: Azure Administrator Associate, Microsoft Certified: Azure Virtual Desktop Specialty, GitHub Foundations, VMware vSphere with Tanzu, Google Cloud Associate Cloud Engineer (previously certified)

Education

Master of Computer Applications (MCA), GITAM Deemed University, India - 2020
Bachelor of Computer Applications (BCA), Tamil Nadu Open University, India - 2017

Languages

English - Professional working proficiency | **German** - A1 certified (TELC, March 2026)